

fNIRS System Optimization for Clinician-Operated Neurofeedback Training

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References



BACKGROUND & MOTIVATION

What is neurofeedback therapy?

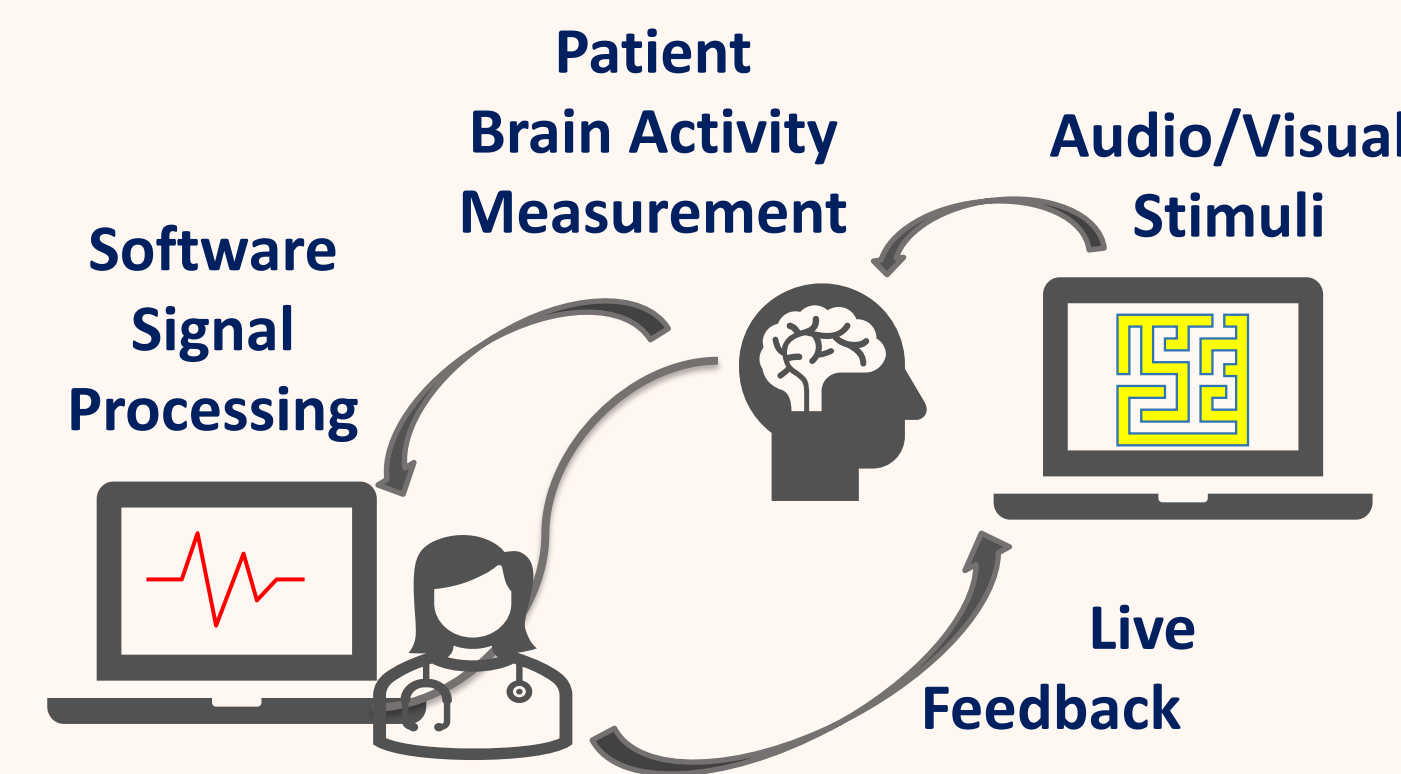
Neurofeedback Therapy (NFT) uses *real-time neuroimaging signals* to provide patients with sensory feedback enabling voluntary control of brain activity.

NFT helps treat patients with:

- Degraded motor function
- Limb loss

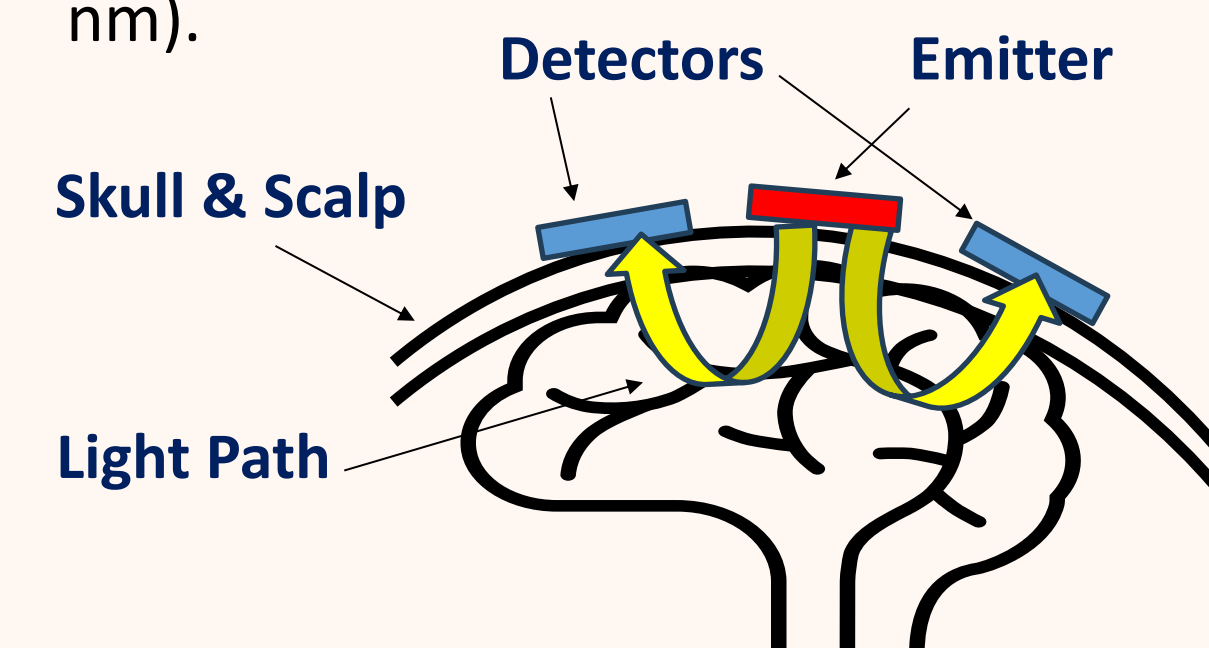
NFT helps improve:

- Limb strength
- Dexterity
- Daily function



What is fNIRS?

Functional Near-Infrared Spectroscopy (fNIRS) is a **non-invasive neuroimaging technique** that measures changes in oxygenation (HbO) and deoxygenation (HbR) hemoglobin using near-infrared light (650 - 900 nm).



Pros:

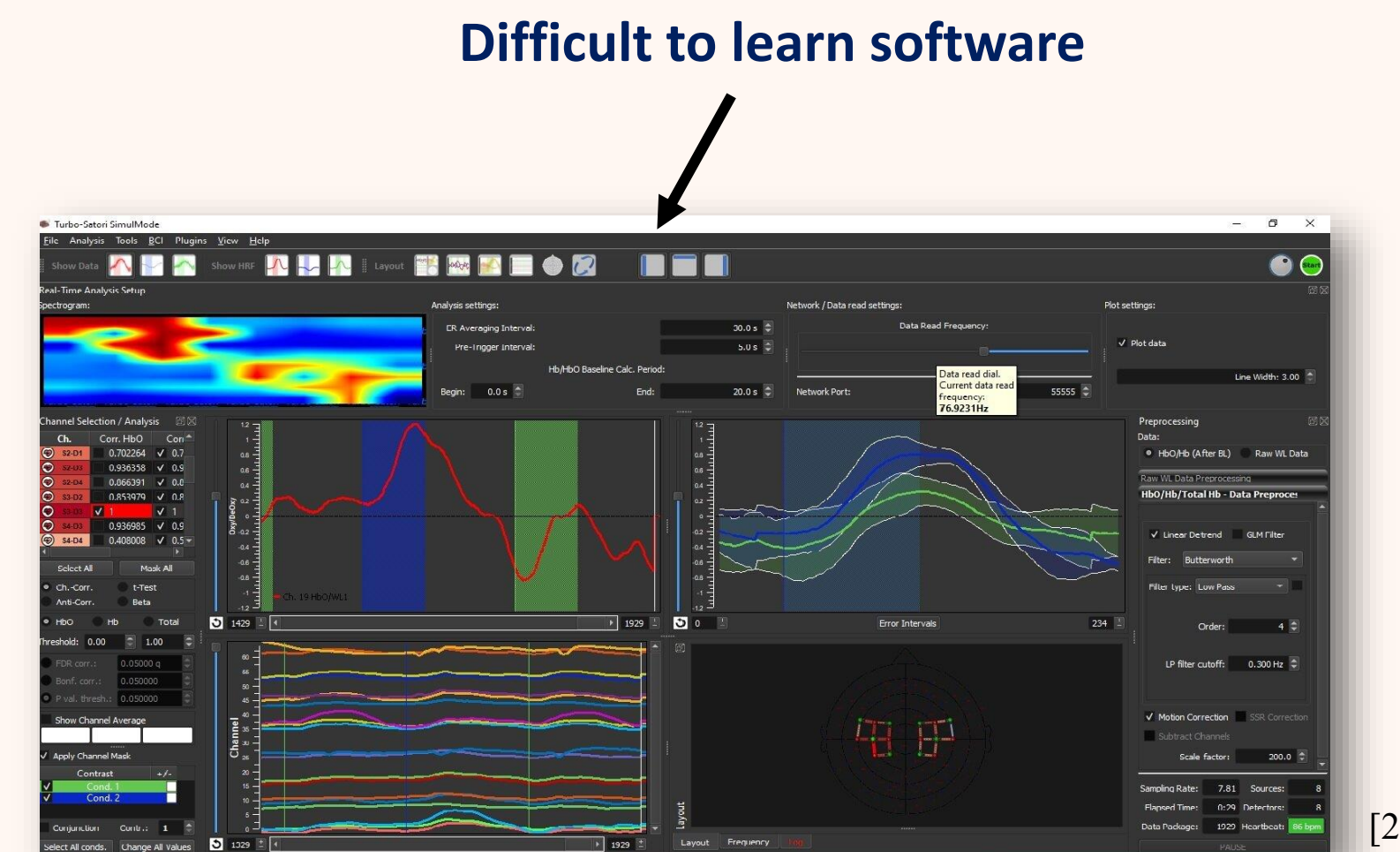
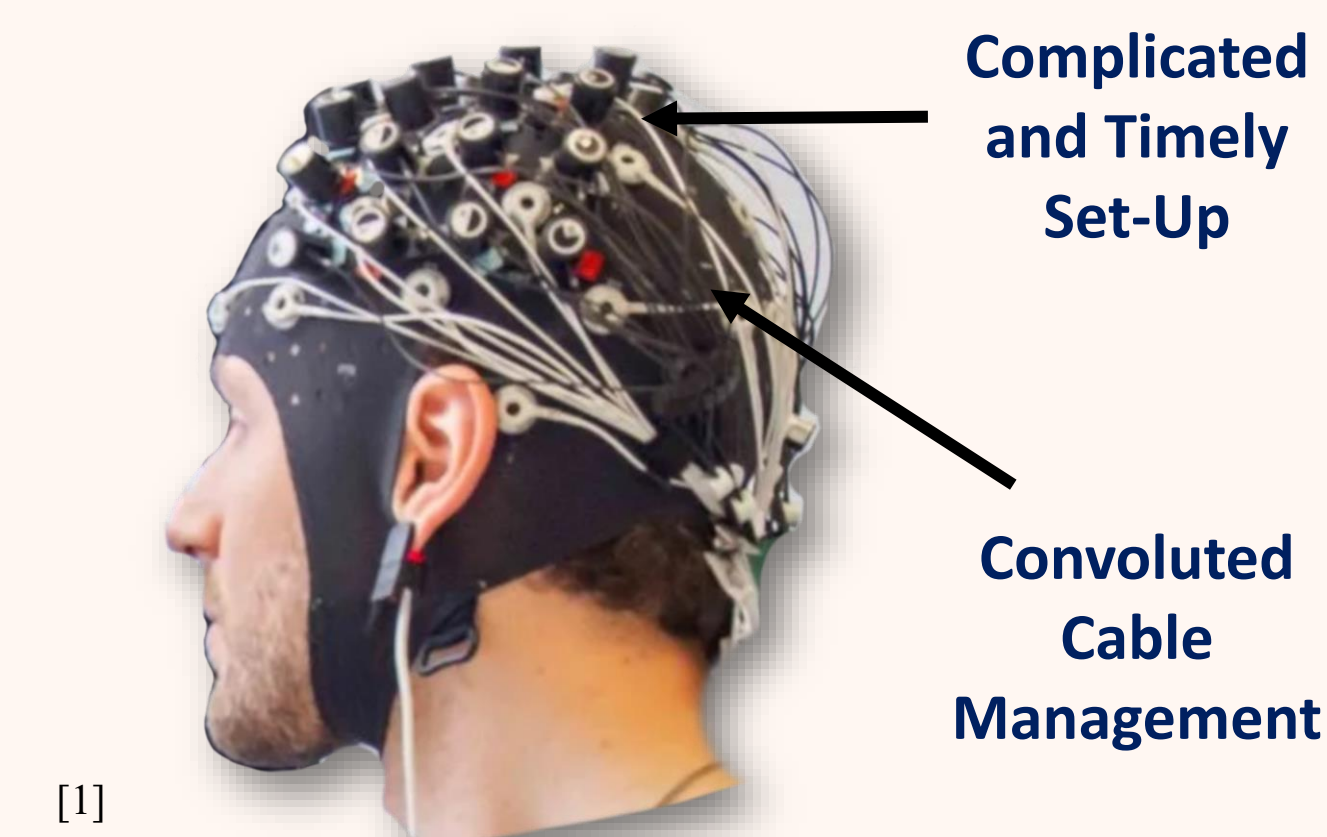
- Non-invasive
- Open source
- Real-time monitoring
- Low cost
- Tolerates movement

Cons:

- Limited spatial resolution
- Slow hemodynamic response
- Long set-up time
- Steep learning curve for operator

Hemodynamic response signals motor cortex activation

Drawbacks of Current Solutions



OBJECTIVES

- Design and assemble a **low-cost fNIRS prototype** (<\$200) focusing on the motor cortex
- Design a simple and polished **UI for clinicians** that requires <1 hour training to successfully master
- Develop an engaging and **gamified patient UI**
- Integrate the full system with **real time signal processing**

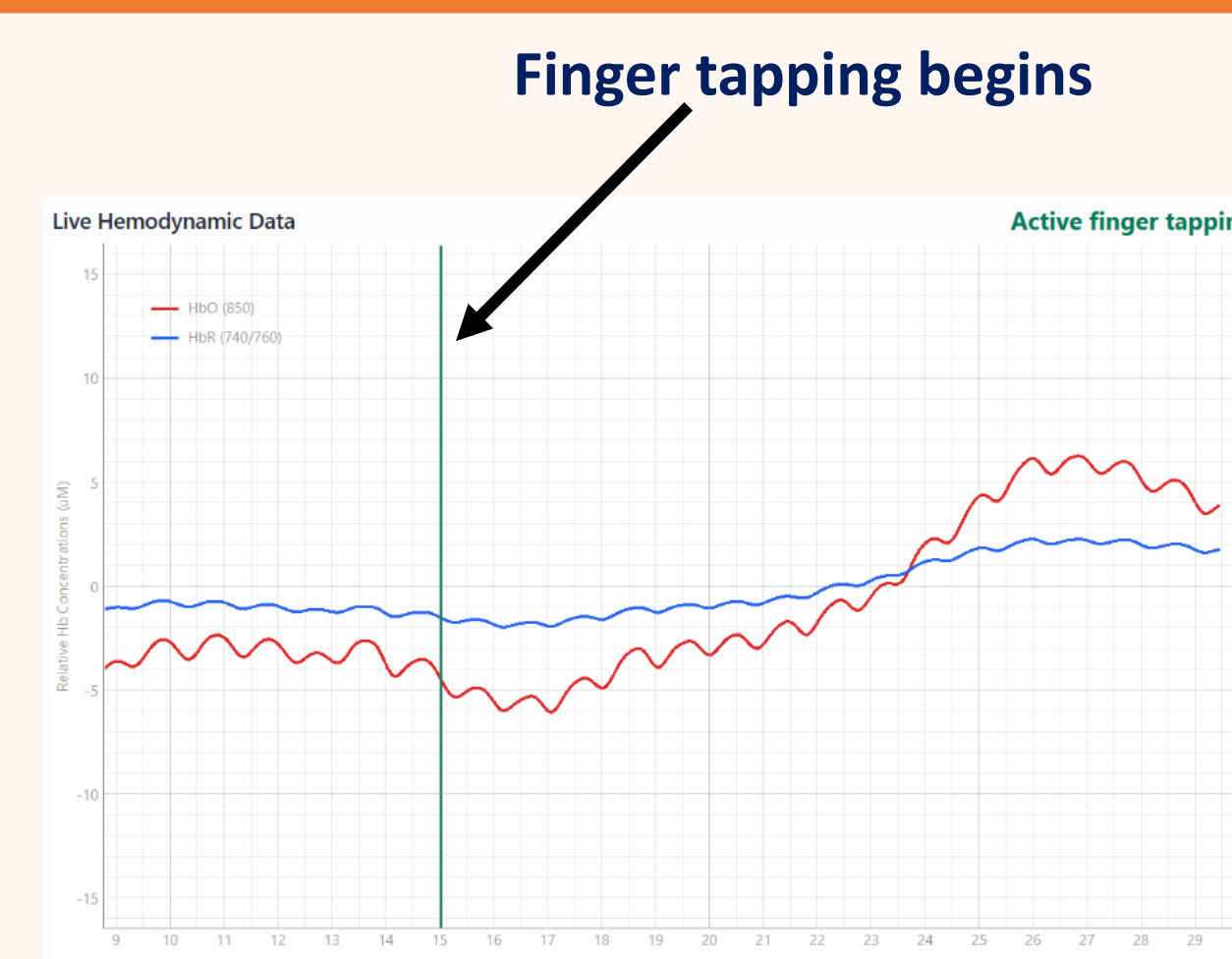
TESTING & RESULTS

Usability Testing

- Hardware and software are **easy to use** with an average training time of under ½ hour for a clinician
- **Patient UI is engaging** with an average engagement rating of 7.8 out of 10

Functionality Testing

- System is **low cost**: \$186.36 CAD
- **Complete and functional data pipeline**: UI receives and displays signals from photodiodes in real-time
- Signal processing successfully **detects and displays motion artifacts** given clear signal data
- 740 nm LED signal to noise ratio does not hit the desired >30dB



OUR SOLUTION

Our low-cost fNIRS system is optimized for clinical integration of upper-limb movement neurofeedback training

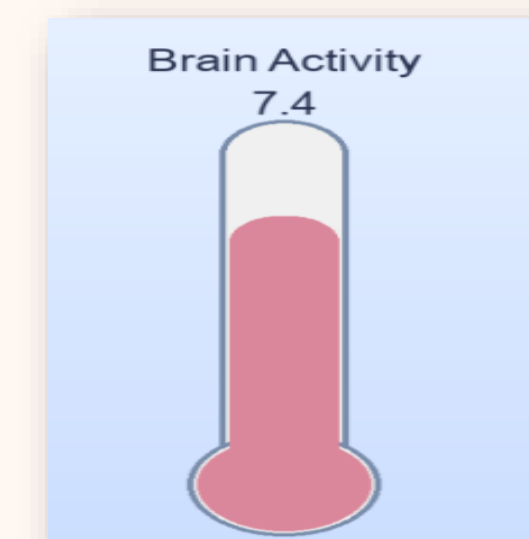


Patient-Facing GUI Games

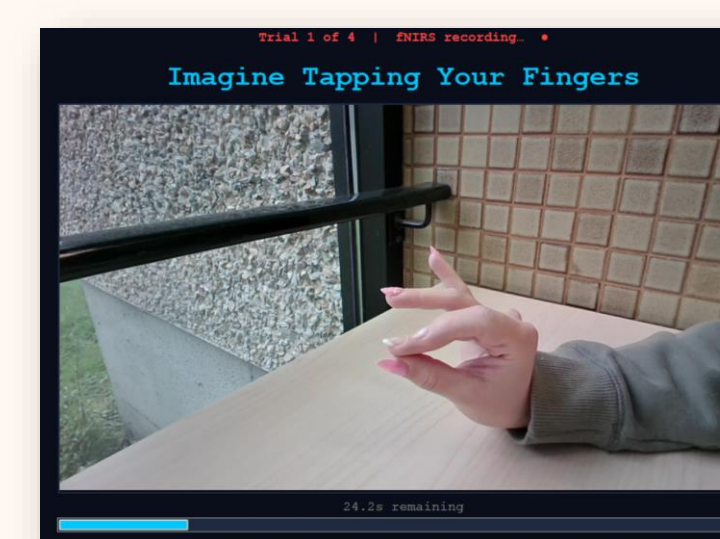
Threshold based block breaking



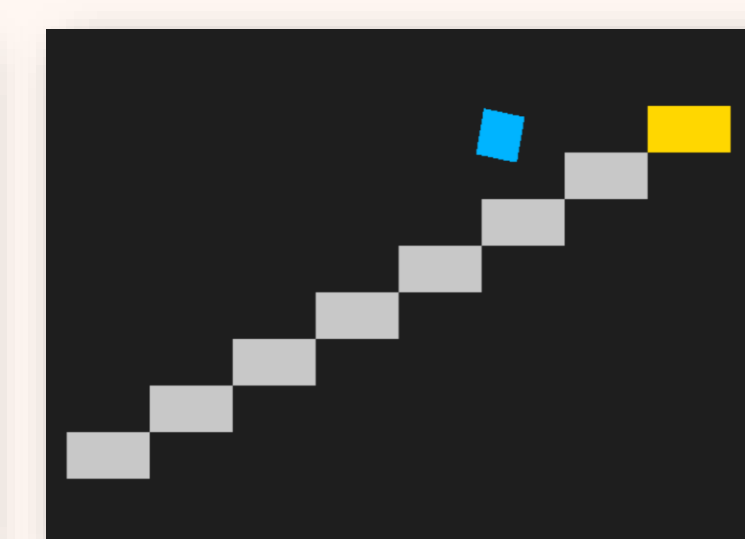
Thermometer rises and falls with brain activity



Visual demonstrations of target movement

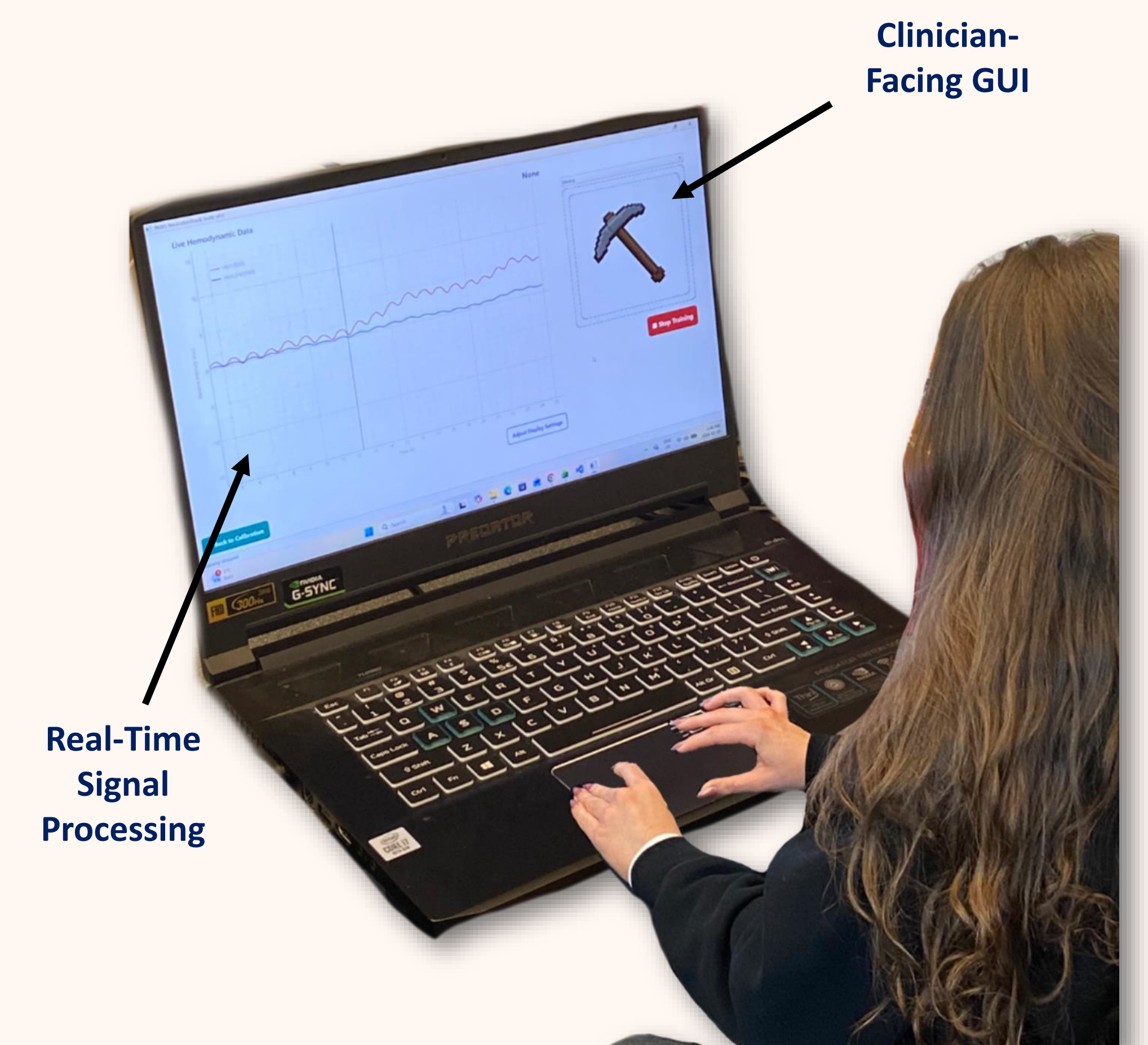


Threshold based block jumping

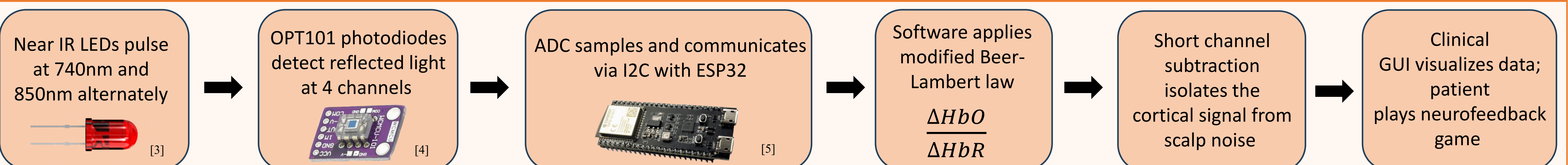


Clinician-Facing GUI Features

- Configuration, calibration, live data processing, and neurofeedback selection and display are all neatly packaged into one executable
- Follows minimalist interface heuristics for a simple, logical workflow



SYSTEM OVERVIEW & DATA FLOW



CONCLUSIONS & FUTURE WORK

Impact

- Makes neurofeedback accessible across diverse economic, geographical, and health populations
- Prototype demonstrates that full research-grade setup is not necessary for clinical NFT
- Low cost enables widespread clinical adoption

Limitations

- High signal to noise ratio from hair and scalp
- Lower power 740nm LED makes scales not comparable between LEDs
- Lower-limb motor cortex detection is out of scope for this prototype

Next Steps

- Iterate signal processing for optimal SNR over scalp and hair
- Design custom PCB
- Explore BLE wireless transmission
- Implement website-based UI
- Open-source release

ACKNOWLEDGEMENTS

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